

Annaluru Sreenadh

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EDUCATION

National Institute of Technology Karnataka

M.Tech Computer Science and Engineering (Information Security) – CGPA: 8.86/10

Surathkal

Sri Venkateswara University College of Engineering

B.Tech Computer Science and Engineering – CGPA: 8.98/10

Aug 2025 – Present

Tirupati, AP

Dec 2021 – May 2025

PROJECTS

Tutor-AI: RAG-based AI Study Assistant

[GitHub](#)

FastAPI, FAISS, PostgreSQL, React

- Built a production RAG pipeline (FAISS + PostgreSQL) with multimodal PDF/PPTX/image ingestion, OCR, and semantic YouTube search across 200+ channels; shipped spaced repetition and adaptive MCQ quizzes with weak-topic detection.
- Engineered secure multi-tenant architecture (JWT auth, bcrypt) with user-scoped FAISS indexes and ownership checks across 40+ API endpoints — production-grade code with end-to-end tests.

StratOS: Autonomous Multi-Agent Competitive Intelligence System

[Live Demo](#) · [GitHub](#)

LangGraph, Qdrant, Groq Llama, LangSmith

- Designed an agent-first orchestration (Scout → Analyst → Strategist → Writer) using LangGraph to automate competitor research and tool use with minimal human input.
- Reduced duplicate signals via Qdrant similarity; reasoned with Groq Llama 3.3 70B and added LangSmith observability; deployed REST API on Render + Docker.

SQL Shield: ML-Based SQL Injection Detector

[Live Demo](#) · [GitHub](#)

FastAPI, Streamlit

- Engineered 98 handcrafted security features from 30K+ SQL queries; achieved 0.995 F1 and 0.9994 ROC-AUC, validated on 30/33 adversarial evasion payloads.
- Deployed a real-time detector with FastAPI, Streamlit dashboard, SHAP explanations, and automated tests to monitor model behaviour under adversarial inputs.

RTP-Streamer: Multi-threaded C++ Media Streaming System

[GitHub](#)

C++17, POSIX Sockets, Multi-threading

- Built a multi-threaded C++17 server/client implementing RFC 3550-style RTP packetization over UDP and an RTSP-lite control protocol (PLAY/PAUSE/TEARDOWN) over TCP; verified byte-for-byte file transfer correctness.
- Designed lock-free synchronization using `std::atomic` and `condition_variable` for zero-CPU-cost pause/resume, and a jitter buffer for out-of-order packet recovery; thread-safety verified via ThreadSanitizer (0 data races).

TECHNICAL SKILLS

Languages & Core: C++, Python, C, SQL, Git

LLMs & Agent Frameworks: LangChain, LangGraph, LangSmith, Prompt Engineering, RAG

Vector & Retrieval: FAISS, Qdrant, embeddings, PostgreSQL

Systems & Networking: Multi-threading, POSIX Sockets, TCP/UDP, RTP/RTSP, ThreadSanitizer

ML & Data Science: XGBoost, Transformers, Scikit-learn, NumPy, Pandas, SHAP

MLOps & Deployment: Docker, MLflow, Render, Streamlit

Web & APIs: FastAPI, REST APIs, React, JWT auth

ACHIEVEMENTS

- GATE CS 2025:** AIR 1476 among 1,70,825 candidates (Top 0.8%)
- Certifications: Introduction to MCP [\[Certificate\]](#) · Claude Code 101 [\[Certificate\]](#)

VOLUNTEERING & COMMUNITY

Main Organizer & Conductor

2025

CuriousClue – Cynosure 2025 (National Tech Fest)

Sri Venkateswara University

- Led end-to-end planning and execution, coordinating communications across 500+ students and faculty for a multi-day national technical fest.